An Evaluation of a Meta-Dimensional Approach to Combine Multimodal Omics and Phenotypic Data to Identify Subgroups of Patients Associated with Morning Cancer-Related Fatigue Severity

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Background and Purpose

Cancer-related fatigue (CRF) is the most common symptom associated with cancer and its treatments. CRF has a negative impact on patients' ability to tolerate treatments and on their quality of life (QOL). Given its high occurrence and negative impact, effective treatments need to be developed for this devastating symptom. One of the major knowledge gaps for CRF is a lack of knowledge of its underlying mechanisms. Increased knowledge of CRF mechanisms could identify potential therapeutic targets.

Integrated Data Analysis

CRF is a complex phenotype, which is the result of environmental (e.g., demographic, clinical, social) and molecular factors. By integrating data from multiple sources, we will have increased power to identify and better interpret the omics-phenotype relationships relative to an analysis that uses only a single source of omics data. We will use this strategy to obtain a more complete biological model of the common and distinct molecular and phenotypic factors associated with the severity of morning CRF.

"Omics" approaches can be used to increase our understanding of the mechanisms that underlie CRF. In systems genomics, either *meta-dimensional* or *multi-staged* data-integrated multi-omics (DIMO) analyses can be used. A meta-dimensional analysis is an approach that combines where all scales of data simultaneously. In contrast, a multi-staged analysis follows a step-wise or hierarchical approach. In the parent study (CA233774), we are performing a multi-staged DIMO analysis (Harris et. al, PMID 33827270) to identify potential mechanisms for CRF (e.g., Kober et. al. 2022, PMID 36373573). In this supplement we are evaluating the utility of a meta-dimensional DIMO approach. Advanced meta-dimensional approaches utilize network analysis methods to integrate multimodal datatypes simultaneously (e.g., patient characteristics and molecular characteristics). This approach will construct integrated joint data profiles of the patients. These profiles will be tested for their association with morning CRF severity. This project complements and enhances the aims of the parent grant by evaluating all data simultaneously (rather than a stepwise approach) and integrates other types of data (i.e., demographic and clinical characteristics). This project is highly economical as it uses the gene expression, methylation, and rich phenotypic data previously or concurrently generated by the previous (CA134900) and parent grant (CA233774).

The purpose of this study was to create a unified fused network using multimodal data types and (1) evaluate this network for joint data profiles of patients, (2) evaluate these profiles for an association with morning CRF, and (3) evaluate these profiles for different demographic and clinical patient characteristics.

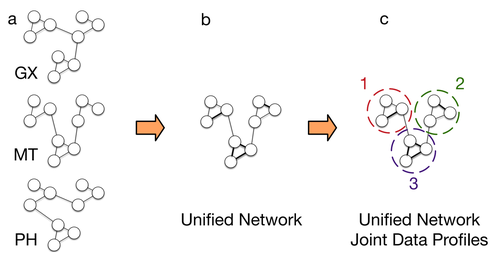

Figure 1. Illustrative example of Similarity Network Fusion (SNF) steps: gene expression (GX), DNA methylation (MT) and phenotype (PH) data sets for the same cohort of patients are used to (a) build patient-by-patient similarity networks. Patients are represented by nodes and patients' pairwise similarities are represented by edges. (b) Network fusion by SNF iteratively updates each of the networks and results in convergence to the final fused unified (UN) network. (c) Spectral clustering is used to identify patients with similar joint data profiles. Adapted from Wang et al. 2014 (PMID 24464287)

Study Sample

Patients and settings

Oncology outpatients (n=1346) who were receiving CTX for breast, lung, gynecological, or gastrointestinal cancers were recruited as part of the previous grant (NCI R01 CA134900). Patients beginning their 2nd or 3rd cycle of CTX were assessed over two complete CTX treatment cycles (i.e., 6 assessments) using the Lee Fatigue Scale (LFS). In the week prior to their 2nd or 3rd cycle of CTX (i.e., Assessment), patients were asked to rate the severity of CRF in the past week upon awakening (morning CRF) and 30 minutes prior to bedtime (evening CRF). Blood samples were drawn for the omics analyses. Differences in phenotypic characteristics between the groups were evaluated using the appropriate parametric and nonparametric tests.

Instruments

Patients completed a demographic questionnaire, Karnofsky Performance Status (KPS) scale, Self-Administered Comorbidity Questionnaire (SCQ), and Alcohol Use Disorders Identification Test (AUDIT). Toxicity of the chemotherapy regimen were rated using the MAX2 index. Medical records were reviewed for disease and treatment information. The 18-item Lee Fatigue Scale (LFS) was used to assess physical fatigue and energy. Each item was rated on a 0 to 10 numeric rating scale. Mean scores were calculated for the 13 fatigue items. Higher scores indicate greater fatigue severity.

Patient Samples

Of the 1346 patients who provided a blood sample, 115 were processed using microarray for gene expression (Illumina HumanHT-12 v4.0 Expression BeadChip) and DNA methylation (Illumina Human Methylation 450K BeadChip) and 291 were processed using RNA-seq for gene expression and microarray for DNA methylation (Illumina Human Methylation EPIC BeadChip). There was no overlap in patients between groups.

Imputation Process

Missing data for demographic and clinical characteristics were imputed by the k-nearest-neighbors method, with k=9 using the entire dataset (n=1346). For continuous variables the Euclidean distance was used to find the nearest neighbors. The imputed value was the weighted average of the nearest neighbors, with each weight originally exp(-dist(x,j)), after which the weights were scaled to one. For categorical variables, distance was 0 if the predictor and the neighbor had the same value and 1 if they did not and the imputed value was the mode of the nearest neighbors.

Pre-Filter of Gene Expression and Methylation Loci

To identify loci to include in the gene expression and methylation networks, differential gene expression and differential methylation analyses were performed for an association with fatigue severity using the smaller group of patients (n=115). Patients were categorized into lower or higher morning CRF using the clinical meaningful cutoff (≥3.2). Differential expression was quantified using empirical Bayes models using edgeR for the RNA-seq data and limma for the microarray data. Cell type deconvolution was performed using the IDOL L-DMR library for cluster of differentiation 8 (CD8) and CD4 T-cells, natural killer (NK) cells, B cells, monocytes, and neutrophils. No differences in cell type estimates was found between patients with lower versus higher levels of morning CRF. Differential analyses were adjusted for age and surrogate variables identified by Surrogate Variable Analysis. Loci having a differential test p-value < 0.05 were included in the subsequent network analysis.

Unified Network Construction with SNF

Gene Expression Data

The larger group of patients (n=291) was used for the network analysis. Of the 16,970 genes evaluated in the RNA-seq dataset, 739 genes were included in the gene expression network. Gene counts were quantified as logCPM and normalized to have mean 0 and standard deviation 1.

Methylation Data

Of the 485,512 probes evaluated in the 450k methylation microarray dataset, 22,360 were included in the methylation network. Methylation levels were quantified as the Beta score and normalized to have mean 0 and standard deviation 1.

Phenotype Data

The phenotype network included two demographic (i.e., age, income) and two clinical (i.e., KPS and SCQ) characteristics. Age and income were selected as demographic characteristics to include in the network due to their association with CRF in previous studies and their potential role in cancer disparities. KPS and SQC were selected as clinical characteristics to include in the network due to their association with CRF and other patient reported outcomes in previous studies. Income was grouped (i.e., >30k, 30-50k, 50-70k, >100k) and treated as an ordinal variable. Phenotype data were normalized to have mean 0 and standard deviation 1.

Unified Network Construction and Clustering

A unified network was created from the three data networks using the Similarity Network Fusion (SNF) method (Wang et al., 2014, PMID 24464287) as implemented in the SNFtool (R package v.2.3.1). First, a similarity network was created for each data type as the pair-wise distance between patients using Euclidean Distance using the default parameters (i.e., number of neighbors for similarity assessment K = 21 (~ participant n/10), scaling factor sigma = 0.5). The unified network was created using the default number of iterations (T = 20).

We identified groups of patients with similar Joint Data Profiles using spectral clustering. Clustering was performed using four different values of k (i.e., k = {2, 3, 4, 5})

Concordance between the cluster assignments made with spectral clustering on the unified network and the three input data networks was evaluated using the Normalized Mutual Information (NMI).

k=2

	UN	GX	MT	PH
UN	1.000	0.008	0.000	0.019
GX	0.008	1.000	0.001	0.000
MT	0.000	0.001	1.000	0.000
PH	0.019	0.000	0.000	1.000

k=3

	UN	GX	MT	PH
UN	1.000	0.075	0.004	0.306
GX	0.075	1.000	0.006	0.005
MT	0.004	0.006	1.000	0.004
PH	0.306	0.005	0.004	1.000

k=4

	UN	GX	MT	PH
UN	1.000	0.076	0.005	0.386
GX	0.076	1.000	0.004	0.019
MT	0.005	0.004	1.000	0.006
PH	0.386	0.019	0.006	1.000

k=5

	UN	GX	MT	PH
UN	1.000	0.057	0.012	0.489
GX	0.057	1.000	0.026	0.012
MT	0.012	0.026	1.000	0.022
PH	0.489	0.012	0.022	1.000

Figure 2. Concordance network NMI scores for the three input data clusters and the fusion cluster for values of k = {2, 3, 4, 5}. Abbreviations: GX, gene expression; MT, methylation; PH, phenotype; UN, unified.

Unified Network

Across the four clustering analysis, the phenotype network had the largest contribution to the unified network (Figure 2). The gene expression data contributed the next most, with the methylation data contributing the least.

There is low concordance network NMI among the individual data networks across all clustering analyses. The different networks contribute unique information.

Joint Data Profiles differ in mean levels of morning fatigue for values of k ≥ 3 (ANOVA; k=2, F=0.53, p-value=0.467; k=3, F=17.52, p-value=6.57×10^{-08}; k=4, F=16.71, p-value=4.95×10^{-10}; k=5, F=13.86, p-value=2.42×10^{-10}).

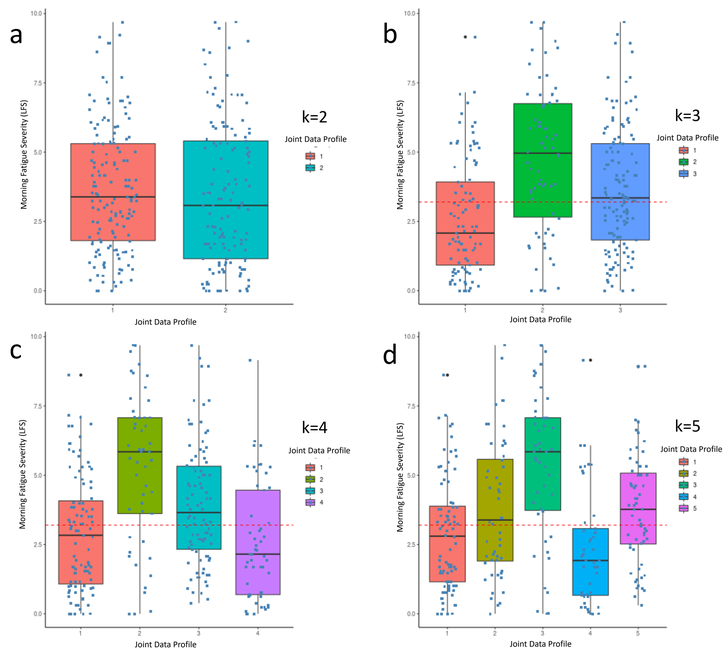

Figure 3. Jitter plots of morning fatigue scores of patients clustered into Joint Data Profiles using spectral clustering of the unified network for gene expression, methylation, and phenotypic data. Four values for k were used: (a) k=2, (b) k=3, (c) k=4, and (d) k=5. The cutoff for clinically meaningful levels of morning fatigue (≥3.2 LFS Score) is shown as a red dashed line. Joint Data Profiles differ in mean levels of morning fatigue for values of k ≥ 3 (k=2, F=0.53, p-value=0.467; k=3, F=17.52, p-value=6.57e-08; k=4, F=16.71, p-value=4.95e-10; k=5, F=13.86, p-value=2.42e-10).

Characterization of Patient Joint Data Profiles (k=3)

Joint Data Patient Profiling

For the groupings assigned using spectral clustering of the fusion network with k=3, patients were evaluated for differences in the phenotypic characteristics used to generate the phenotype network (Table 1). As compared to Patients in Profile 3, patients in Profiles 1 and 3 were older and reported lower performance. As compared to Patients in Profile 3, patients in Profiles 1 and 3 reported lower income and a lower comorbidity burden.

Table 1. Differences in Demographic, Clinical Characteristics Among the Joint Data Patient Profiles for Variables Included in Phenotype Network in the SNF Analysis and Spectral Clustering (k=3)

Characteristic	Profile 1 33.3% (n=97)	Profile 2 22.7% (n=66)	Profile 3 44.0% (n=128)	Statistics
	Mean (SD)	Mean (SD)	Mean (SD)	
Age (years)	62.6 (9.1)	61.6 (10.4)	49.8 (11.0)	F = 52.25, p < 2×10^{-16} 1 and 2 > 3
Karnofsky Performance Status score	86.5 (8.9)	73.4 (10.4)	73.0 (12.2)	F = 50.03, p < 2×10^{-16} 1 and 2 < 3
Self-administered Comorbidity Questionnaire score	4.6 (2.3)	10.3 (3.2)	4.0 (1.5)	F = 185.7, p < 2×10^{-16} 1 < 2, 2 < 3
	% (n)	% (n)	% (n)	
Annual household income				X^2 = 43.62, p = 8.78×10^{-8}
Less than $30,000+	16.5 (16)	47.0 (31)	10.2 (13)	1 < 2, 2 > 3
$30,000 to $70,000	16.5 (16)	22.7 (15)	21.1 (27)	NS
$70,000 to $100,000	26.8 (26)	10.6 (7)	23.4 (30)	NS
Greater than $100,000	40.1 (39)	19.7 (13)	45.3 (58)	1 > 2, 2 < 3

Abbreviations: SD = standard deviation

Differences in Patient Demographic and Clinical Characteristics

Tests for differences in patient demographic and clinical characteristics are reported in Table 2. Patient reported different levels of fatigue across Profiles (ANOVA F = 17.52, p = 6.57×10^{-8}). Patients in Profile 2 reported higher levels of morning fatigue as compared to Profiles 1 and 3. Patients in Profile 2 had fewer years of education as compared to Profiles 1 and 3. Patients in Profile 2 a higher body mass index and more likely to have a diagnosis of lung cancer as compared to Profile 1 and 3.

Of note, patients did not differ among groups on self reported gender or ethnicity. In addition, patients did not differ among groups for hemoglobin, hemocrit, MAX2, or exercise.

Table 2. Differences in Demographic, Clinical Characteristics Among the Joint Data Patient Profiles.

Characteristic	Profile 1 33.3% (n=97)	Profile 2 22.7% (n=66)	Profile 3 44.0% (n=128)	Statistics
	Mean (SD)	Mean (SD)	Mean (SD)	
Education (years)	16.5 (3.0)	14.6 (3.0)	16.3 (3.0)	F = 10.95, p = 2.6×10^{-5} 1 > 2, 2 < 3
Body mass index (kg/m^2)	25.5 (4.9)	29.0 (7.0)	25.1 (5.2)	F = 11.31, p = 1.87×10^{-5} 1 < 2, 2 > 3
Alcohol Use Disorders Identification Test score	2.8 (2.1)	2.9 (2.0)	2.7 (2.0)	F = 0.40, p = 0.670
MAX2 score	0.17 (.08)	0.18 (0.9)	0.19 (0.07)	F = 1.53, p = 0.218
Hemoglobin	11.6 (1.3)	11.4 (1.6)	11.4 (1.4)	F = 1.06, p = 0.347
Hemocrit	34.9 (3.7)	34.2 (4.8)	34.2 (4.0)	F = 1.04, p = 0.355
Morning Fatigue	2.6 (2.1)	4.8 (2.6)	3.7 (2.3)	F = 17.52, p = 6.57×10^{-8} 1 < 2 and 3, 2 > 3
	% (n)	% (n)	% (n)	
Gender (% female)	70.1 (68)	75.8 (50)	81.3 (104)	X^2 = 3.80, p = .149
Self-reported ethnicity				
White	36.7 (68)	20.0 (37)	43.2 (80)	X^2 = 8.13, p = .229
Asian or Pacific Islander	22.4 (11)	22.5 (11)	55.1 (27)	
Black	37.5 (9)	33.3 (8)	29.2 (7)	
Hispanic, Mixed, or Other	27.2 (9)	30.3 (10)	42.4 (14)	
Married or partnered (% yes)	62.9 (61)	51.5 (34)	65.6 (84)	X^2 = 3.778, p = .151
Lives alone (% yes)	27.8 (27)	33.3 (22)	14.1 (18)	X^2 = 11.03, p = .004 2 > 3
Child care responsibilities (% yes)	10.3 (87)	15.2 (56)	32.8 (42)	X^2 = 18.60, p = 9.17×10^{-5} 1 < 3
Elder care responsibilities (% yes)	3.1 (3)	13.6 (9)	6.3 (8)	X^2 = 6.96, p =.031 NS
Past or current history of smoking (% yes)	37.1 (36)	43.9 (29)	23.4 (30)	X^2 = 9.65, p = .008 NS
Exercise on a regular basis (% yes)	70.1 (68)	60.6 (40)	68.8 (88)	X^2 = 1.18, p = 0.404
Cancer diagnosis				X^2 = 32.47, p = 1.33×10^{-5}
Breast cancer	33.0 (32)	19.7 (13)	45.3 (58)	2 < 3
Gastrointestinal cancer	43.3 (42)	33.3 (22)	39.8 (51)	NS
Gynecological cancer	17.5 (17)	25.8 (17)	11.7 (15)	NS
Lung cancer	6.2 (6)	21.2 (14)	3.2 (4)	1 < 2, 2 > 3
Prior cancer treatment				X^2 = 17.67, p = .007
No prior treatment	(19)	(16)	(43)	NS
Only surgery, CTX, or RT	(36)	(26)	(61)	NS
Surgery and CTX, or surgery and RT, or CTX and RT	(24)	(15)	(14)	1 > 2
Surgery and CTX and RT	(18)	(9)	(10)	NS

Abbreviations: CTX = chemotherapy, kg = kilograms, m^2 = meters squared, NS = not significant, RT = radiation therapy, SD = standard deviation

Discussion

This study was the first to generate a unified fused network from gene expression, methylation and phenotype data for patients undergoing CTX. We evaluated the SNF approach to identify profiles of patients that were associated with morning fatigue severity. We found that Joint Data Profiles differ in mean levels of morning fatigue for values of k ≥ 3, highlighting the utility for identifying groups of patients with different experiences of morning fatigue during CTX.

Limitations & Future Directions

This analysis requires independent validation.

Future research is needed evaluate the characteristics of groups of patients identified from k=4 and k=5 spectral clustering, calculate the relative contributions of the features in driving the patient groups, evaluate additional datatypes (e.g., genotype, other lab values, imaging), and evaluate for differences between profiles for molecular markers (e.g., genotype, gene expression, metabolomics, DNA methylation).

In terms of the methodology, future analyses should perform hyperparameter optimization (Markello et al., 2021, PMID 33402689), evaluate for model stability of cluster membership (e.g., n=100 resample 80% of patients) and reliability of top characteristics (e.g., n=100 random draws of 50% patients) (Stefanik et al., 2018, PMID 29105664), and evaluate other methods (i.e., Chi square distance) to generate the patient similarity network for discrete data types (i.e., income).

Conclusions

This approach has demonstrated the potential to identify clinically informative groups of patients through the fusion of different data types. These profiles can be useful to identify high risk patients and novel therapeutic targets. These methods may be generalizable to other symptoms and patient-reported outcomes associated with cancer and its treatment (e.g., sleep disturbance, depression, anxiety, dyspnea, pain).

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